

Deep Learning

Mental Health Assistant

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April 10th



Idea & Motivation

Why this project?

- Rise in interest in psychology and therapy
- Mental health services are overloaded and often unaffordable
- Information online is scattered and unreliable

Solution:

A chatbot that answers mental health questions using trusted scientific books

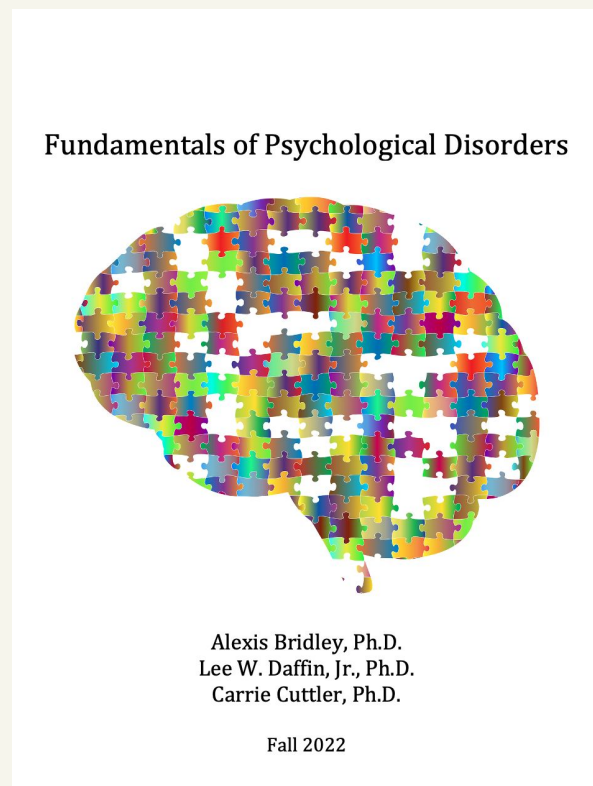
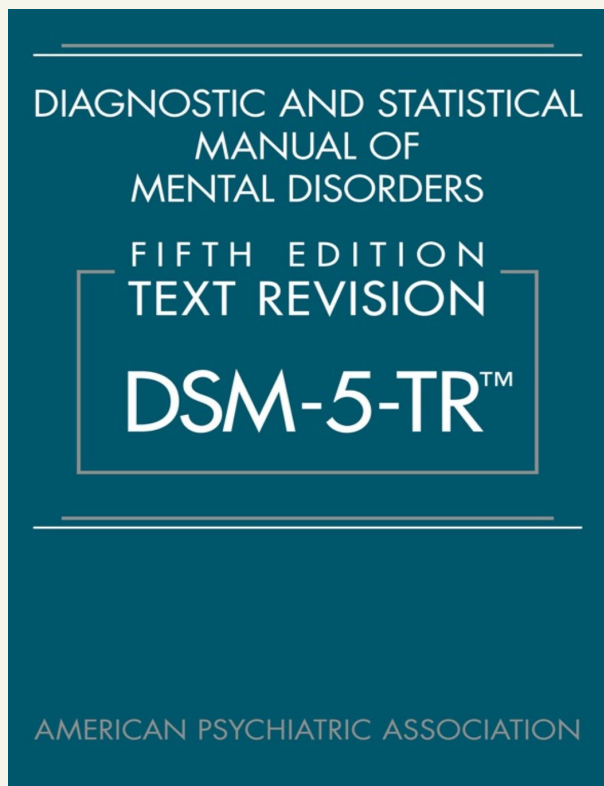


What is the goal?

Build a chatbot that:

- Understands user questions
- Finds relevant content from mental health scientific textbooks
- Generates clear, safe, and accurate responses





Why these sources:

- Scientifically accurate
- Structured content (e.g., disorders, symptoms, treatments)
- Ideal for structured semantic search

Milvus and Hugging Face Integration

- Vector DB: Milvus for fast retrieval of semantically similar chunks
- Embedding model: all-MiniLM-L6-v2
- Chosen LLMs:
 - deepset/roberta-base-squad2 (Extractive QA)
 - deepset/roberta-large-squad2 (Extractive QA)
 - google/flan-t5-large (Generative QA)
 - mistralai/Mistral-7B-Instruct-v0.1 (Generative QA)

Model Selection Process

Evaluation Criteria:

- Relevance of answers
- Completeness
- Understandability

Methods used:

- Sample question set (ex: "What is bipolar disorder?")
- Compare output of each model

1. [deepset/roberta-base-squad2](#)

- "What is a bipolar disorder?"

Output: 'cyclothymic disorder'

- "What are the signs of burnout, and how is it different from regular stress?"

Output: 'ineffective coping mechanisms'

- "Lately, I feel exhausted and unmotivated even after a full night's sleep. What could this mean?"

Output: 'these symptoms are a component of the disorder'

- "How can I stop overthinking everything? It's ruining my focus and peace of mind."

Output: 'what if something happens'

- "What's the difference between a psychologist and a psychiatrist?"

Output: 'physician, psychiatrist, and psychologist'

- What criteria are used to define abnormal behavior in psychology?

Output: 'abnormal behavior'

- How is Major Depressive Disorder diagnosed, and what are its primary symptoms?

Output: 'substantial changes in affect, cognition, and neurovegetative functions'

- What distinguishes Post-Traumatic Stress Disorder (PTSD) from Acute Stress Disorder?

Output: 'cannot be comorbid disorders'

- Can you describe the different types of dissociative disorders and their characteristics?

Output: 'dissociative identity disorder, dissociative amnesia'

- How do Obsessive-Compulsive Disorder (OCD) and related disorders manifest in individuals?

Output: 'negative thoughts or anxieties related to an unpleasant event'

- What are the diagnostic criteria for Schizophrenia Spectrum Disorders?

Which model did I choose?

google/flan-t5-large - the Generative QA model.

Why does it perform better?

- Full-sentence, paraphrased, user-friendly answers.
- Handles nuance and long context better.
- Can adapt the tone (e.g., calm, professional, supportive) with prompt tuning.
- Balance between quality and Colab compatibility
- Better than extractive QA in providing helpful explanations

FLAN-T5,
a yummy model
superior to **GPT-3**



Critical Evaluation

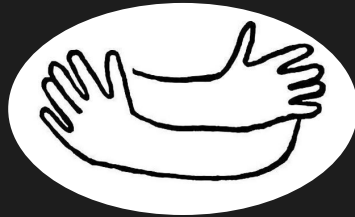
Strengths:

- Fast, accurate answers
- Grounded in real, scientific knowledge
- 2 strong scientific books
- Customizable for other books/domains

Limitations:

- Requires Colab GPU to run fluently, re execution of the code, several google accounts
- Sensitive to prompt quality
- Not a substitute for professional help (ethical disclaimer)

Demo



Potential Improvements

- Add support for:
 - Multiple sources (books, research papers)
 - Multilingual questions
 - Sentiment detection
- Fine-tune for more empathy and tone control
- Deploy as web app available for others to use



Thank You!

Any Questions?

